

$$r = c\Delta t/\Delta x, \quad 0 < r \leq 1,$$

$$u_j^{n+1} = 2u_j^n - u_j^{n-1} + r^2(u_{j-1}^n - 2u_j^n + u_{j+1}^n),$$

$$u_j^1 = u_j^0 + \Delta t g_j + \tfrac{1}{2}r^2(u_{j-1}^0 - 2u_j^0 + u_{j+1}^0),$$

$$g_j = u_t(x_j, 0) = 0, \quad u_0^n = u_N^n = 0,$$

$$\text{error} = O(\Delta t^2 + \Delta x^2) \quad \text{for smooth solutions.}$$